Topological Energy Condensate Theory (TECT):

Research Programme Roadmap — Projects I through VI
(Documents Math01–Math36)

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Abstract

Topological Energy Condensate Theory (TECT) derives gravity, gauge fields, and Standard Model fermions from a single $\mathbb{C}^3$-valued energy condensate whose equilibrium configuration is Body-Centered Cubic (BCC). Documents Math01–Math36 decompose into six sequential projects: (I) BCC ground-state theorem via $\mathbb{Z}_2$-symmetric Landau–Brazovskii theory ($N_{\text{loop}} = 6$ for BCC vs. 2 for FCC and 0 for SC); (II) emergent $U(1)$ gauge structure; (III) topological Weyl/Dirac fermion emergence; (IV) microscopic Dirac carrier acceptance (Clifford closure, Gates 1–4); (V) three-generation flavor structure and SM+GR infrared limit; (VI) numerical PDE certification on the Class II branch at $(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62)$.

1 Overview: TECT Logical Chain and Project Map

The fundamental hypothesis is

$$\boxed{\text{Universe} = \text{BCC condensate } \Psi_0 \xrightarrow{\text{fluctuations}} \text{gauge fields} + \text{fermions} + \text{gravity}.} \tag{1}$$

The condensate field $\Psi(x) \in \mathbb{C}^3$ evolves under

$$\mathcal{L}_{\text{mic}} = \mathcal{L}_{\Psi}^{\text{core}}[\Psi] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[\Psi], \tag{2}$$

with core operator

$$\mathcal{L}_{\Psi}^{\text{core}} = -\mu^2\Psi + Z\nabla^2\Psi - Y\nabla^4\Psi - \lambda|\Psi|^2\Psi - \gamma|\Psi|^4\Psi. \tag{3}$$

Project Status Summary

Project	Files	Objective	Status
I	Math01–05	BCC ground-state uniqueness	Proved
II	Math06–09	Emergent $U(1)$ gauge structure	Proved
III	Math10–14	Topological Weyl/Dirac fermions	Proved
IV	Math15–24	Microscopic Dirac carrier audit	Certificate pending
V	Math25–30	Flavor structure & SM+GR limit	Framework complete
VI	Math31–36	Numerical PDE certification	Active frontier

Logical dependency chain:

$$\text{I} \longrightarrow \text{II} \longrightarrow \text{III} \longrightarrow \text{IV} \longrightarrow \text{V} \longrightarrow \text{VI}. \tag{4}$$

2 Project I: BCC Ground-State Theorem (Math01–05)

2.1 Core Objective

Prove that the BCC Bravais lattice is the unique global minimum of the $\mathbb{Z}_2$-symmetric Landau–Brazovskii free energy under pure quartic (4-wave) fluctuations at 1-loop.

2.2 Governing Free Energy

$$\begin{aligned} F[\{\Psi_{\mathbf{k}}\}] = &\sum_{|\mathbf{k}|=k_0} \left[r + \kappa(k^2 - k_0^2)^2\right]|\Psi_{\mathbf{k}}|^2 \\ &+ u \sum_{\substack{\mathbf{k}_1+\cdots+\mathbf{k}_4=0 \\ |\mathbf{k}_i|=k_0}} \Psi_{\mathbf{k}_1}\Psi_{\mathbf{k}_2}\Psi_{\mathbf{k}_3}\Psi_{\mathbf{k}_4}, \end{aligned} \tag{5}$$

with $\mathbb{Z}_2$: $\Psi \to -\Psi$ forbidding cubic invariants.

2.3 Main Theorem

Theorem 2.1 (BCC Selection — Math01)**.** *The non-trivial 4-wave resonance loop multiplicities are*

$$N_{\text{loop}}^{\text{BCC}} = 6, \quad N_{\text{loop}}^{\text{FCC}} = 2, \quad N_{\text{loop}}^{\text{SC}} = 0. \tag{6}$$

At 1-loop, $\Delta F_{\text{BCC}} < \Delta F_{\text{FCC}} < \Delta F_{\text{SC}}$, and BCC is selected via a fluctuation-induced first-order transition.

2.4 Robustness Extensions

Math02 (Universality A) Finite shell thickness $\Delta k/k_0 \ll 1$ preserves $N_{\text{loop}}^{\text{BCC}} > N_{\text{loop}}^{\text{FCC}}$.

Math03 (Universality B) Amplitude variation $A_i = \bar{A} + \delta A_i$ shows equal-amplitude BCC is a strict local minimum.

Math04–05 (Multi-Shell) For $S = \bigcup_{a=1}^{K} S_a$, BCC selection extends under the single-shell dominance condition.

2.5 Deliverables

1. Exact N_{loop} for BCC, FCC, SC. [Done]
2. 1-loop ordering $\Delta F_{\text{BCC}} < \Delta F_{\text{FCC}}$. [Done]
3. Universality under Δk and amplitude variation. [Done]
4. Multi-shell resonance network theorem. [Done]

3 Project II: Emergent Gauge Structure (Math06–09)

3.1 Core Objective

Derive emergent $U(1)$ and higher gauge fields from low-energy fluctuations around the BCC background Ψ_0.

3.2 Transverse Decomposition

Expanding $\Psi = \Psi_0 + \delta\Psi$ and decomposing for each $\mathbf{k}$:

$$\delta\mathbf{u} = u_L\hat{\mathbf{k}} + u_1\mathbf{e}_1 + u_2\mathbf{e}_2. \tag{7}$$

3.3 Main Theorems

Theorem 3.1 ($U(1)$ Emergence — Math06)**.** *In the IR limit $|\mathbf{k}|a \ll 1$, the transverse sector carries $SO(2) \simeq U(1)$. The emergent covariant derivative is $D_\mu \sim \partial_\mu + iA_\mu$, $A_\mu \in \mathfrak{u}(1)$.*

Theorem 3.2 ($\mathbb{CP}^2$ Bundle — Math07)**.** *The orientation $z(x) \in \mathbb{C}^3$, $z^\dagger z = 1$, defines a principal $U(1)$ bundle over $\mathbb{CP}^2$. The Berry connection yields $F = dA$.*

Theorem 3.3 (Dirac Coupling Non-Vanishing — Math09)**.** *For any $G_* \neq 0$, the reduced shell coefficients $v_\alpha = e_i^\alpha(\mathsf{N}_{ij})^a(G_*)_j \neq 0$, $\alpha = 1, 2$.*

3.4 Deliverables

1. Emergent $U(1)$ from transverse BCC modes. [Done]
2. $\mathbb{CP}^2$ geometric bundle. [Done]
3. Non-vanishing Dirac coupling. [Done]
4. Full $SU(3) \times SU(2) \times U(1)$ from $\mathbb{C}^3$ bundle. [In progress]

4 Project III: Topological Fermion Emergence (Math10–14)

4.1 Core Objective

Establish massless Weyl/Dirac fermions at Bloch-band crossings on the BCC reciprocal shell, protected by residual valley symmetry.

4.2 Valley-Paired Block

$$H_D(\mathbf{p}) = \begin{pmatrix} h_+(\mathbf{p}) & 0 \\ 0 & h_-(\mathbf{p}) \end{pmatrix}, \quad h_-(\mathbf{p}) = -h_+(-\mathbf{p}). \tag{8}$$

4.3 Main Theorems

Theorem 4.1 (Weyl Node — Math10, Math11)**.** *After shell-adapted coordinate change:*

$$H_D(\mathbf{p}) = \sum_{i=1}^{3} v_i p_i \alpha_i, \quad \alpha_i := \tau_z \otimes \sigma_i. \tag{9}$$

Theorem 4.2 (Mass Protection — Math12)**.** *The residual $\mathbb{Z}_2^{\text{valley}}$ forbids constant mass operators:*

$$\det H_D(0) = 0. \tag{10}$$

Theorem 4.3 (Little-Group Mismatch — Math13)**.** *Inequivalent representations of H_* in $\mathcal{H}_D$ and $\mathcal{H}_R$ imply $V(0) = 0$ (no remote-sector contamination at $p = 0$).*

Theorem 4.4 (Clifford Closure — Math14, Math18)**.** *The full low-energy space $\mathcal{H}_{\text{low}} = \mathbb{C}^2_{\text{int}} \otimes \mathbb{C}^2_{\text{valley}}$ realises a minimal 4×4 Dirac representation.*

4.4 Deliverables

1. Weyl node at BCC shell crossing. [Done]
2. Mass protection via $\mathbb{Z}_2^{\text{valley}}$. [Done]
3. Little-group mismatch $\Rightarrow V(0) = 0$. [Done]
4. Clifford closure on $\mathbb{C}^4$. [Done]

5 Project IV: Microscopic Dirac Carrier Acceptance (Math15–24)

5.1 Core Objective

Finite operator audit confirming all four acceptance gates:

$$\boxed{\text{Gate 1: isolation} \ \wedge \ \text{Gates 2–3: spectrum} \ \wedge \ \text{Gate 4: mass protection.}} \tag{11}$$

5.2 Step-by-Step Programme

Math15 (Audit Formulation). Open questions recast as finite operator audit on the linearized Bloch operator around the locked branch. Central object: $P_0 = z_0 z_0^\dagger$.

Math16–17 (PDE Status).

$$\boxed{\text{proof chain complete;} \quad \text{microscopic coefficient certificate: pending.}} \tag{12}$$

Math18 (Clifford Closure). The reduced symbol on the doubled valley space yields the 4×4 Dirac Hamiltonian. $\mathrm{Cl}(3,0)$ closes on $\{\tau_z \otimes \sigma_i\}_{i=1}^{3} \cup \{\tau_x \otimes \sigma_0\}$.

Math19–20 (Explicit Shell Basis).

$$\mathcal{H}_D^{(N)} \cong \mathrm{span}\{|G_*, \sigma, \tau\rangle\}_{\sigma,\tau \in \{+,-\}}. \tag{13}$$

Math21 (Acceptance Milestone). Frontier identified as Gates 2–3 (spectrum audit) plus Gate 4 (mass protection).

Math22–23 (Non-Enriched Benchmark). $\det H^{(1)}_{\text{non-enr}}(0) \neq 0$ verified for $N = 1$.

Math24 (Small-p Drift Control).

$$\|H^{(N)}(p) - H^{(N)}(0)\| \leq C_{\text{drift}} \cdot |p|, \quad |p| \leq \rho. \tag{14}$$

5.3 Deliverables

1. Gates 1–4 proved. [Done]
2. Explicit coefficient certificate. [Pending — Project VI]

6 Project V: Flavor Structure and SM+GR Infrared Limit (Math25–30)

6.1 Core Objective

Establish the three-generation left-handed space $\mathcal{H}_L$ and the minimal theorem checklist for an SM+GR-type infrared description.

6.2 Main Results

Math25 (Local Uniqueness). The 3×3 Hessian of the reduced potential is positive-definite at the selected branch: local uniqueness established.

Math26 (SM+GR Target Theorem).

$$\boxed{\mathrm{TECT}_{\mathrm{IR}} \supseteq \mathrm{SM} + \mathrm{GR}.} \tag{15}$$

Minimal checklist: 7 required lemmas for $SU(3) \times SU(2) \times U(1)$ minimally coupled to $g_{\mu\nu}$.

Math27 (Three-Generation Isolation, K1).

$$\mathcal{H}_L = \mathrm{Ran}(P_L), \quad \dim \mathcal{H}_L = 3. \tag{16}$$

Math28 (Kinetic Gram Matrix, K2A).

$$G_{IJ} = \langle L_I | \hat{K} | L_J \rangle, \quad I, J = 1, 2, 3. \tag{17}$$

Must be positive-definite for physical normalizability.

Math29 (BCC Ground State).

$$\Psi_0(\mathbf{x}) = \phi_0 \sum_{\mathbf{G}\in\mathcal{S}_{\mathrm{BCC}}} z_{\mathbf{G}}\, e^{i\mathbf{G}\cdot\mathbf{x}}, \quad |\mathcal{S}_{\mathrm{BCC}}| = 12. \tag{18}$$

Math30 (Completion Status).

$$\boxed{\text{proof complete;} \quad \text{certificate incomplete (microscopic audit required).}} \tag{19}$$

6.3 Deliverables

1. $\dim \mathcal{H}_L = 3$ (three generations). [Done]
2. SM+GR checklist framework (7 lemmas). [Done]
3. Explicit G_{IJ} entries. [Pending — Project VI]

7 Project VI: Numerical PDE Certification (Math31–36)

7.1 Core Objective

Complete the finite microscopic audit and certify the PDE coefficient extraction on the Class II working branch.

7.2 Locked Parameters

$$\boxed{(\mu^2, \lambda, \gamma) = (0.26,\ -0.43,\ 1.62).} \tag{20}$$

7.3 Sequential Steps

Math31 (Enlarged Quartic Kernel). Four-class quartic kernel beyond the failed (u, v, κ) truncation.

Math32 (Residual Quartic Closure). Exact four-class upgrade:

$$F_4 = u_1\rho_1 + u_2\rho_2 + u_3\rho_3 + u_4\rho_4. \tag{21}$$

Math33–34 (PDE Coefficients + Seed Profile).

$$\lambda_\parallel = \tfrac{\lambda_c}{2} I_3, \quad \alpha = \beta = -\frac{\beta_X \gamma_\perp}{M_X^2} I_c. \tag{22}$$

Explicit seed profile inserted; Bloch matrix elements $\mathcal{U}_j(0)$ extracted.

Math35 (Canonical Longitudinal Coefficient).

$$u_\parallel = \frac{1}{\phi_0}\, \hat{e}_\parallel^\dagger\, \mathcal{U}(0)\, \hat{e}_\parallel. \tag{23}$$

Math36 (Class II Elimination — Active Frontier). Heavy mediator X_i^a integrated out:

$$X_i^a = -\frac{1}{M_X^2}\big(\alpha_X J_i^a + \beta_X K_i^a\big). \tag{24}$$

The resulting effective Lagrangian is the starting point for full BCC linearization.

7.4 Deliverables

1. Four-class quartic kernel. [In progress]
2. PDE coefficients $(\lambda_{\parallel}, v_{\perp}, v_{\parallel})$. [In progress]
3. Positive-definite G_{IJ} with explicit values. [Target]
4. Full BCC linearization after Class II elimination. [Target]
5. $m^* = 0.3138$ derived analytically. [Final target]

8 Immediate Next Step: Math37 Target

Five sequential steps to close Project VI:

1. Insert Eq. (24) into $\mathcal{L}_{\text{mic}}$ to obtain the exact induced effective quartic Lagrangian.
2. Perform BCC Bloch decomposition on the 12-vector star $\mathcal{S}_{\text{BCC}}$.
3. Extract $(v_{\parallel}, v_{\perp}, \lambda_{\parallel})$ numerically at parameters (20).
4. Verify $\det H_D(0) = 0$ (Weyl node survives Class II elimination).
5. Compute m^* analytically and verify $m^* = 0.3138 \pm \delta m^*$ against numerical simulation.

Open Problem 8.1 (Analytical Derivation of m^*)**.** *Derive $m^* = 0.3138$ analytically from the BCC linearization of the Class II effective Lagrangian at $(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62)$. This constitutes the first analytical confirmation of a TECT-predicted physical observable against numerical simulation.*

9 Overall Programme Status

$$
\begin{aligned}
&\text{Projects I–III: all theorem-level proofs complete.}\\
&\text{Project IV: Gates 1–4 proved; explicit coefficient values pending.}\\
&\text{Project V: framework complete; explicit Gram matrix entries pending.}\\
&\text{Project VI: active frontier. Math36 (Class II elimination) is the immediate next step.}
\end{aligned}
\tag{25}
$$